



Bayesian quantile-based correction and synthesis for climate products

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QUESTION

Correct first, then synthesize

Can a Bayesian quantile model learn source-specific corrections from paired gauge observations and retrospective product archives, propagate them into new ensemble forecasts, and synthesize one posterior predictive distribution on operational update timescales?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge on the $\log(1 + x)$ scale. Before each forecast origin, observed flow is paired with retrospective ECMWF/GloFAS and NOAA/NWS products to learn dynamic source corrections. After the origin, issued ensembles and staged exogenous covariates carry those corrections into source-adjusted quantile forecasts and one posterior predictive distribution.



USGS observations

15-min public gauge record aggregated to daily log-flow; target for state updating and held-out verification.



ECMWF GloFAS

retrospective alignment and daily 51-member forecast guidance to 28 days.



NOAA NWS

retrospective alignment and hourly forecasts evaluated on common daily leads 1–8.

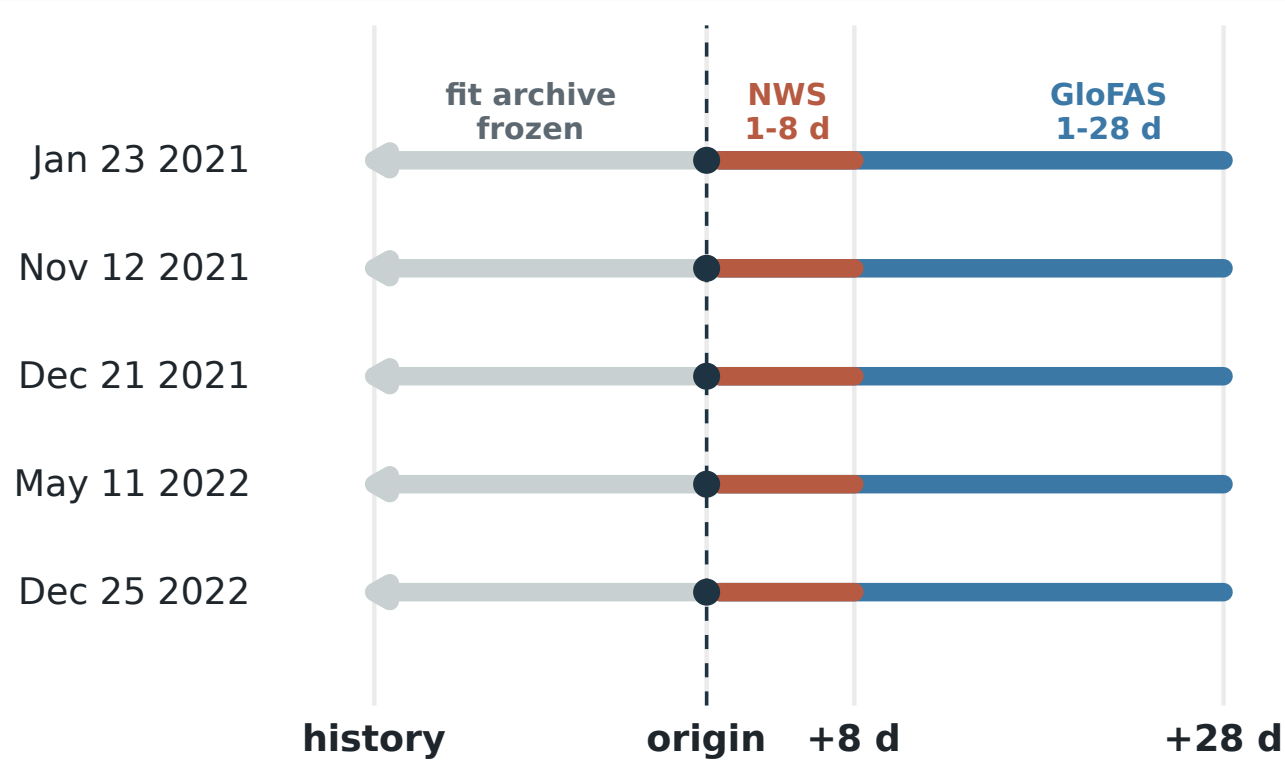


Exogenous information

Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION

Rolling-origin holdout design



Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE

Scalable Bayesian quantile fitting

Seven independent extended Dynamic Quantile Linear Model (exDQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. The fitted quantile forecasts are checked for crossing in the application and combined into the posterior predictive distribution scored by CRPS.

7

quantile lanes

≤

12,995

USGS archive days

~

2–4 h

fit + synthesis

Parallel quantile lanes

Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

DQLM baseline versus exDQLM

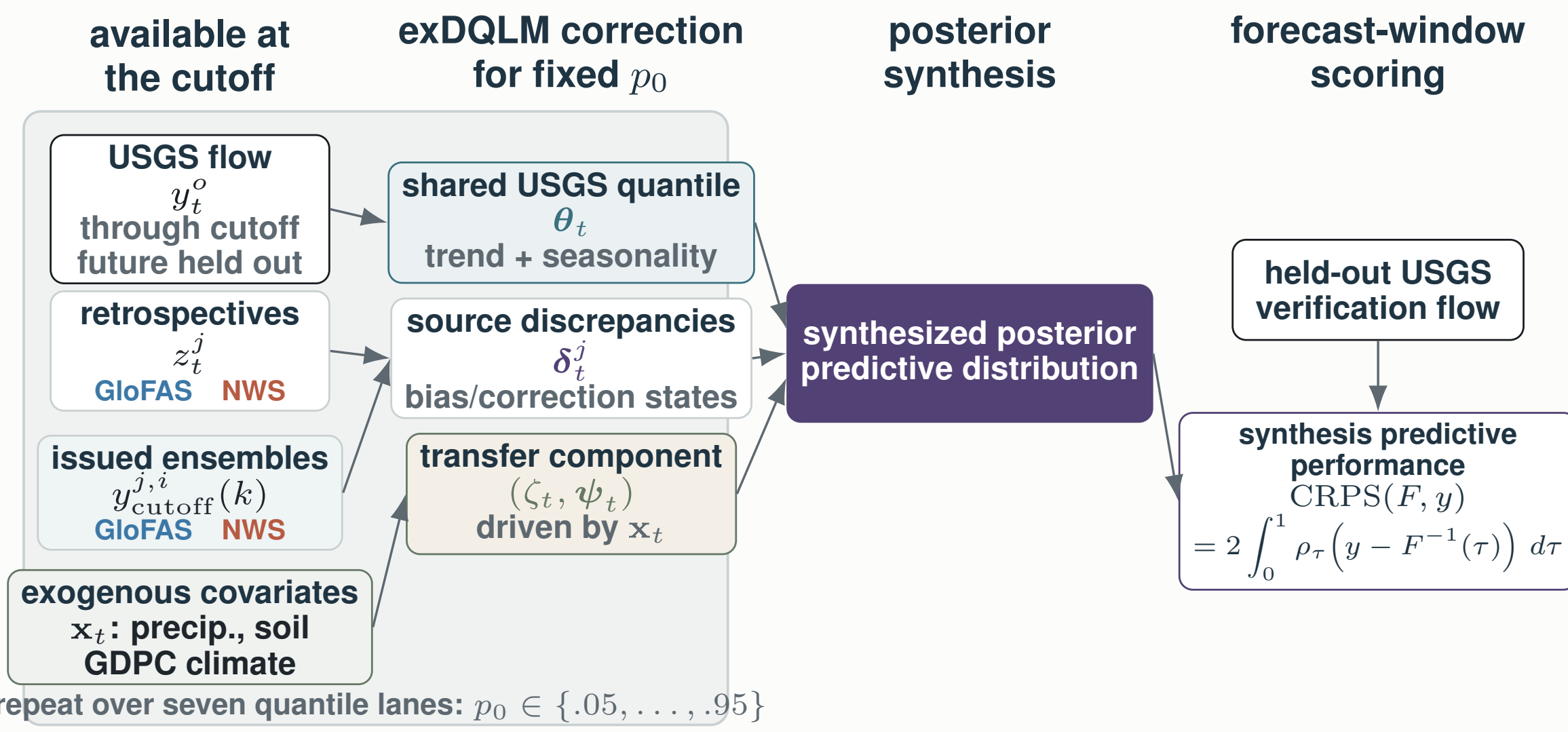
The asymmetric-Laplace DQLM is conjugate under its latent representation; the selected exDQLM replaces AL by exAL for source-specific scale and skewness flexibility.

Scalable VB-LD updates

Mean-field variational Bayes keeps Kalman filtering and backward smoothing for Gaussian state blocks; VB-LD uses local modes, curvature, and Delta-method moments for non-conjugate exAL terms.

MODEL

Main Workflow



Source-aware exDQLM workflow. For each target quantile p_0 , pre-origin USGS observations and retrospective product histories identify a latent river-flow quantile and source-specific correction states. After the cutoff, issued ensemble forecasts and staged exogenous covariates propagate corrected forecast-window quantiles. The seven fitted quantile lanes are then synthesized into one posterior predictive distribution and scored only with held-out USGS observations.

Dynamic Quantile Linear Model (exDQLM)

For each target quantile p_0 , let T denote the forecast origin/cutoff. The selected exDQLM links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Source discrepancies learn pre-origin corrections; forecast ensembles and staged covariates then propagate corrected quantiles after the origin.

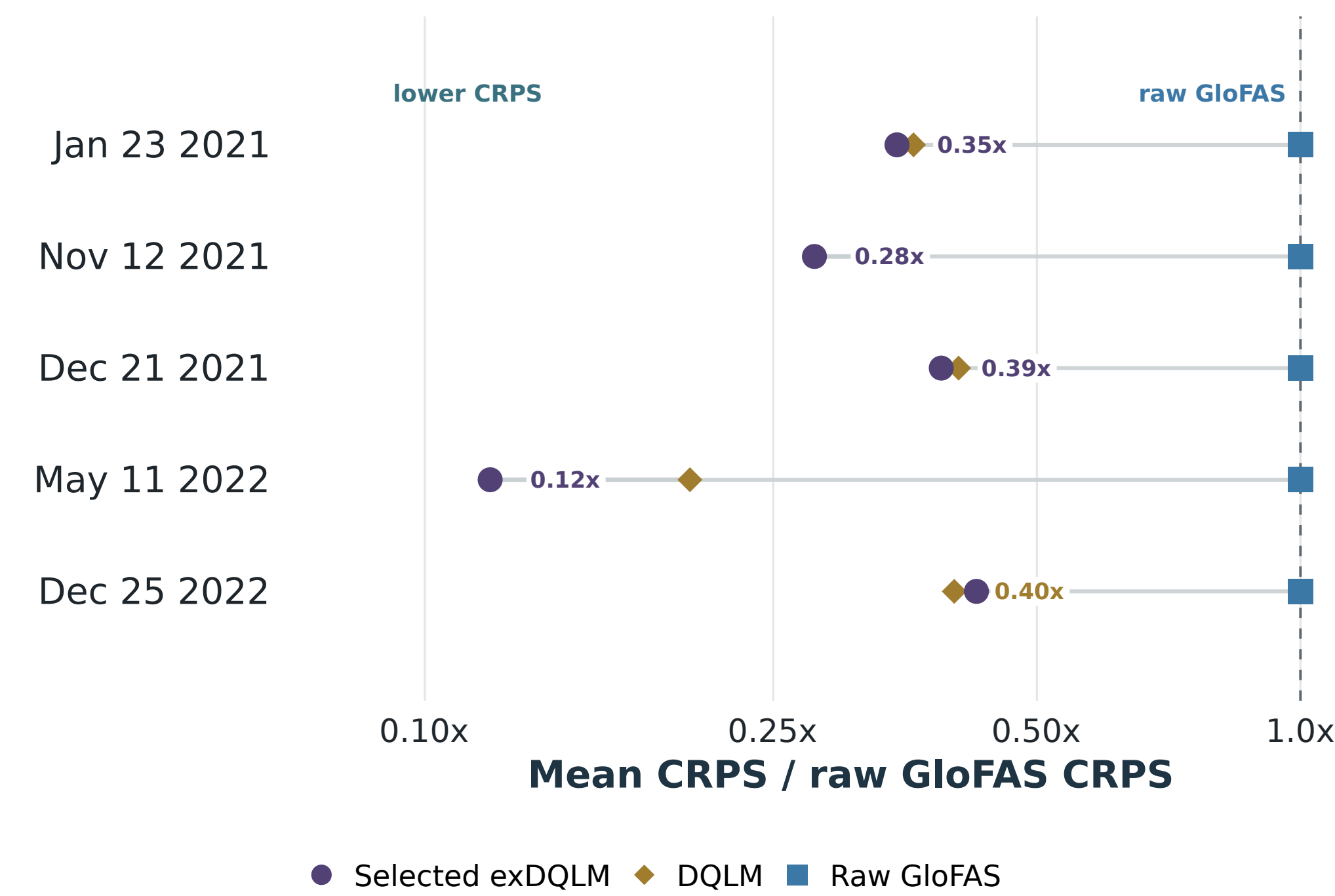
$$\begin{aligned} y_t^o &\sim \text{exAL}_{p_0}(\mathbf{F}_t' \boldsymbol{\theta}_t + \zeta_t, \sigma^o, \gamma^o), & 1 \leq t \leq T, \\ z_t^j &\sim \text{exAL}_{p_0}(\mathbf{F}_t' \boldsymbol{\theta}_t + \boldsymbol{\delta}_t^j + \zeta_t, \sigma^j, \gamma^j), & j \in \mathcal{J}, 1 \leq t \leq T, \\ y_{T+k}^{j,i} &\sim \text{exAL}_{p_0}(\mathbf{F}_{T+k}'(\boldsymbol{\theta}_{T+k} + \boldsymbol{\delta}_{T+k}^j + \zeta_{T+k}, \sigma^j, \gamma^j), & j \in \mathcal{J}, i \in \mathcal{I}_j, k \in \mathcal{K}_j. \\ \boldsymbol{\theta}_t &| \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t \boldsymbol{\theta}_{t-1}, \mathbf{W}_t^\theta), & \boldsymbol{\delta}_t^j | \boldsymbol{\delta}_{t-1}^j \sim \mathcal{N}(\mathbf{G}_t \boldsymbol{\delta}_{t-1}^j, \mathbf{W}_t^{\delta^j}). \\ \zeta_t &| \zeta_{t-1}, \boldsymbol{\psi}_{t-1} \sim \mathcal{N}(\lambda \zeta_{t-1} + \mathbf{x}_t' \boldsymbol{\psi}_{t-1}, w_\zeta^2), & \boldsymbol{\psi}_t | \boldsymbol{\psi}_{t-1} \sim \mathcal{N}(\boldsymbol{\psi}_{t-1}, \mathbf{W}_t^\psi). \end{aligned}$$

Here $\boldsymbol{\theta}_t$ is the shared trend–seasonal river quantile, $\boldsymbol{\delta}_t^j$ is the source-specific correction, and $(\zeta_t, \boldsymbol{\psi}_t)$ is the exogenous-covariate block driven by $\mathbf{x}_t$ (local precipitation, local soil moisture, and GDPC climate summaries). The DQLM comparator uses the conjugate AL likelihood; the selected exDQLM uses exAL and updates non-conjugate σ, γ terms with VB-LD.

RESULTS

1–28-step-ahead forecast CRPS comparison

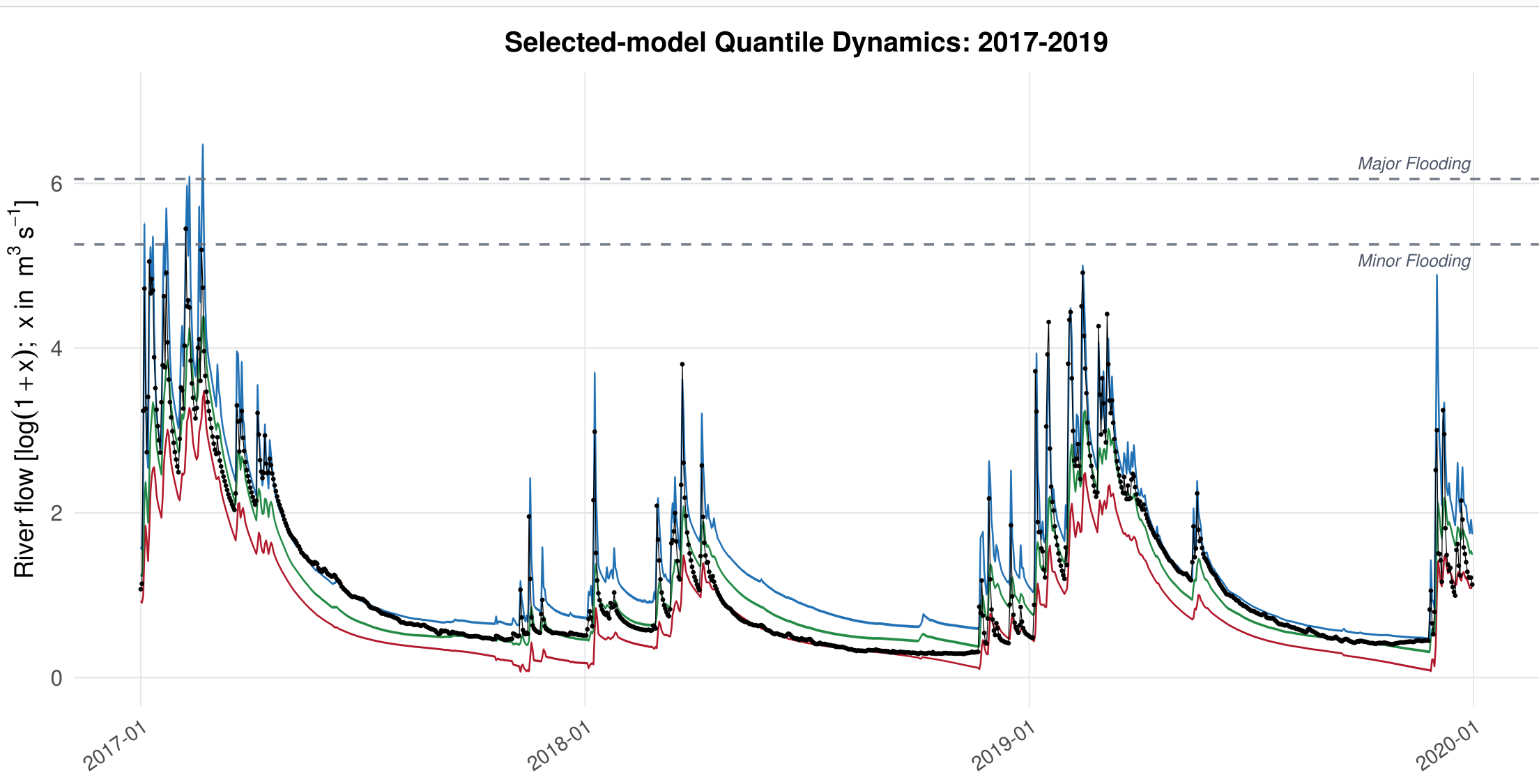
Selected exDQLM is lowest at four of five origins; DQLM leads at 2022-12-25



CRPS is the continuous ranked probability score; lower values are better. DQLM is the conjugate AL-likelihood baseline, while selected exDQLM denotes the exAL extension. Raw NWS is evaluated separately over common leads 1–8, not mixed into this 1–28 lead GloFAS-supported benchmark.

FIT

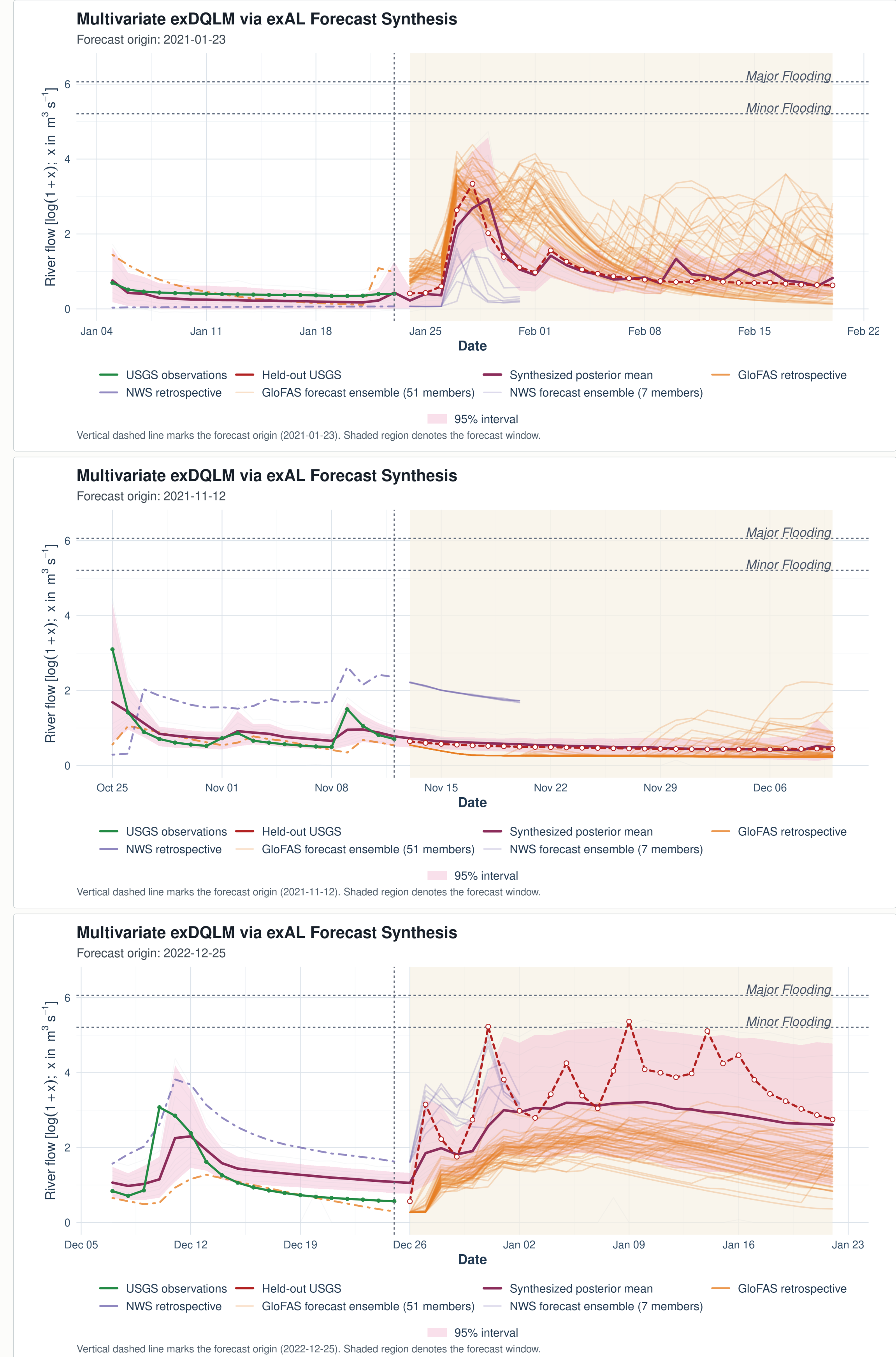
Wet-period quantile fit



Historical fit diagnostic. Selected exDQLM fitted quantile locations during the 2017–2019 wet period. The 5th, 50th, and 95th quantile fits are shown with observed USGS flow; wider upper-tail bands make winter high-flow behavior visible.

EXAMPLE

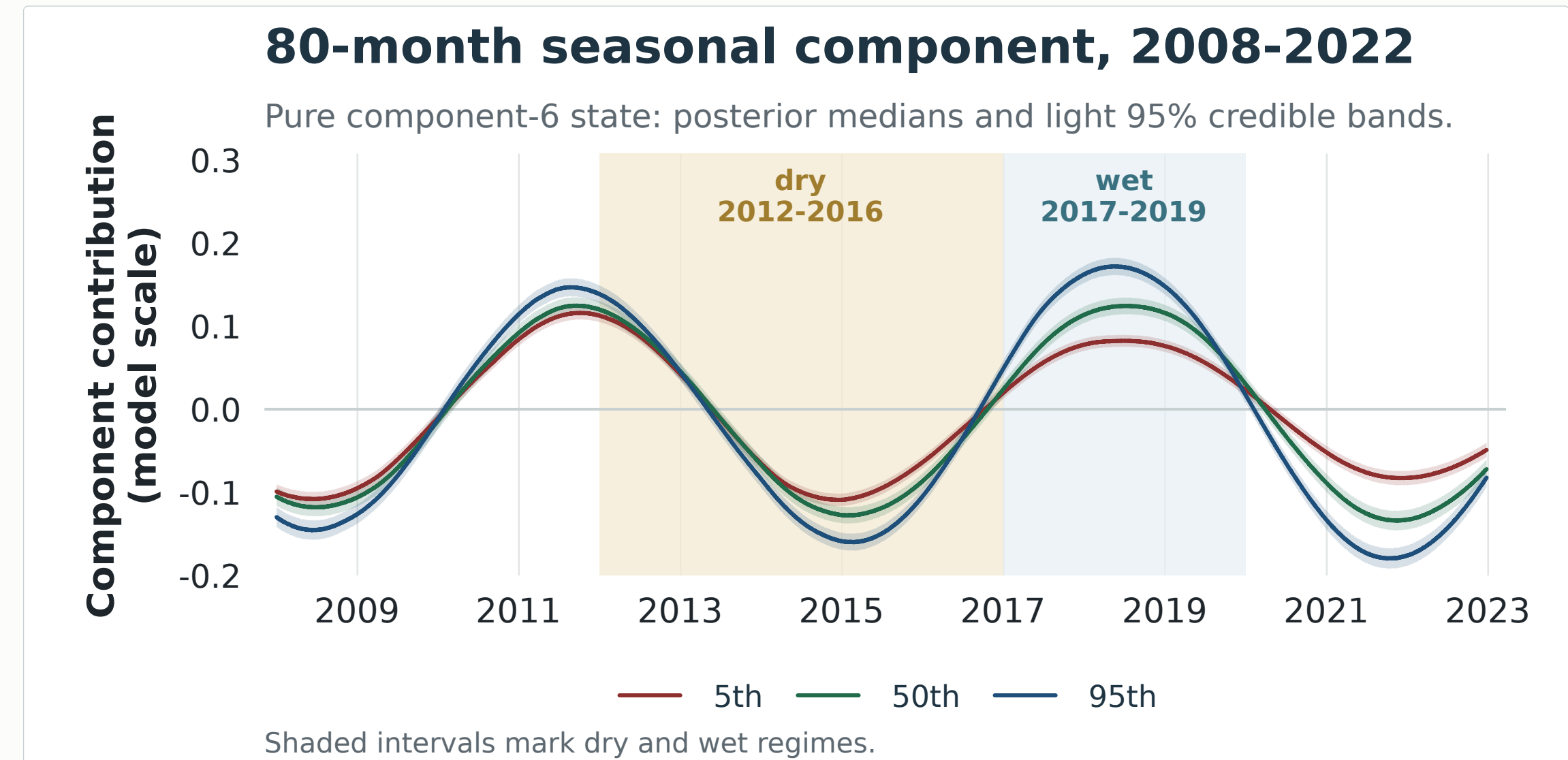
Three forecast-window examples



Origin-level illustrations. Three selected exDQLM forecast windows show how corrected quantile forecasts combine into a posterior predictive envelope after the cutoff. The panels are illustrative checks of behavior at individual origins; the aggregate comparison remains the five-origin CRPS benchmark.

DYNAMICS

80-month seasonal component



Retrospective diagnostic. From the selected exDQLM at the 2022-12-25 origin. Lines show the isolated 80-month state component, without the trend component added, for the 5th, 50th, and 95th target-quantile lanes over 2008–2022; light ribbons show 95% credible bands and shaded intervals mark dry and wet regimes.

TAKEAWAYS

Main takeaways

- Selected exDQLM has the lowest 1–28-step-ahead CRPS at four of five origins; DQLM leads at 2022-12-25.
- Historical product archives learn source-specific corrections that propagate into new forecast ensembles.
- Seven corrected quantile lanes synthesize one posterior predictive distribution; NWS is compared separately over common leads 1–8.

